

**FLUKE**

Certificate of Calibration

Everett Service Center**Certificate Number:** EVL1056908**Status:** AS-LEFT : Pass**Date of Calibration:** 29 May 2025**Date Due:** 29 May 2027**Manufacturer:** Fluke**Temperature:** 22.6 °C**Model:** 5522A**Relative Humidity:** 46.4 %**Serial Number:** 2806902**Issue Date:** 08 Jul 2025**Option(s):** PQ/SC600**Description:** Calibrator**Procedure:** Fluke 55XX Mainframe

Rev 20240708

Customer: MODERN SURVEYING CALIBRATION & TESTING LABS(FZE)**Location:** ERBIL

IQ

Asset ID: MSL/E/001**PO Number:** PO1250013**RMA/SO Number:** 33033490

This calibration is traceable to the International System of Units (SI) through recognized national metrology institutes (NIST, NRC, PTB, NPL, etc.), radiometric techniques, or natural physical constants and is in conformance with ISO/IEC 17025:2017. Calibration certificates without identification of the authorizing person are not valid. This certificate applies to only the item identified and shall not be reproduced except in full, without the specific written approval by Fluke Corporation. The calibration interval (date due) is the responsibility of the end user.

This certificate of calibration may contain data that is not covered by the Scope of Accreditation. The unaccredited measurement points are indicated by the # symbol or confined to clearly marked sections.

Measurement uncertainties at the time of calibration are given where applicable. They are calculated in accordance with the method described in the ISO Guide to the Expression of Uncertainty in Measurement (GUM). The reported expanded uncertainty of measurement is stated as the combined standard uncertainty of measurement multiplied by the coverage factor k such that the coverage probability corresponds to approximately 95 % and $k=2$.



Authorized By
Lori Howard
Calibration Technician

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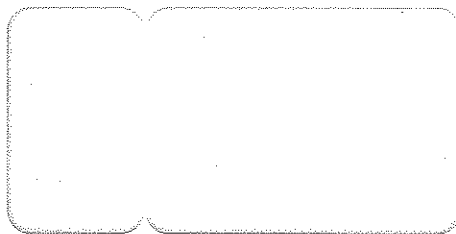
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This calibration has been completed in accordance with:
 The Fluke Corporate Quality Manual, Revision 129
 The Fluke 17025 Quality Manual, Revision 009

Statements of conformity are made in this certificate as described in Fluke document FCM 7008.1, paragraph 2a revision 002

The following describes the symbols found in the data tables that follow:

No symbol: Measurement value is less than maximum permissible error.

P* : Pass Conditional (error less than maximum permissible error but 2% PFA not guaranteed)

F : Fail (error greater than maximum permissible error)

: Out of accredited scope

Standards Used

Asset	Description	Cal Date	Cal Due
18101	Clarke-Hess 6000A Phase Meter	30 Jun 2024	30 Jun 2025
21098	Fluke 1504 Thermometer Readout	07 Jan 2025	07 Jan 2026
21122	Fluke 5640 Thermistor Probe	30 Jul 2024	30 Jul 2025
21071	Fluke 5790B AC Measurement Standard	27 Aug 2024	27 Aug 2025
20720	Fluke 742A-19M Resistance Standard	11 Nov 2024	11 Nov 2025
21165	Fluke 742A-1M Resistance Standard	30 Aug 2024	30 Aug 2025
18296	Fluke 7526A Precision Process Calibrator	15 Oct 2024	15 Jul 2025
21189	Fluke 8588A Reference Multimeter	19 Sep 2024	19 Sep 2025
16955	Fluke A40B-10mA Precision Current Shunt	17 Oct 2024	17 Oct 2025
16957	Fluke A40B-20A Precision Current Shunt	17 Oct 2024	17 Oct 2025
16959	Fluke A40B-2A Precision Current Shunt	17 Oct 2024	17 Oct 2025
16960	Fluke A40B-500mA Precision Current Shunt	17 Oct 2024	17 Oct 2025
16958	Fluke A40B-50mA Precision Current Shunt	17 Oct 2024	17 Oct 2025
16954	Fluke A40B-5A Precision Current Shunt	17 Oct 2024	17 Oct 2025
21185	Fluke FTE-190M Standard Resistor	13 Mar 2025	13 Jul 2025
20823	Fluke FTE-300M Standard Resistor	07 Nov 2024	07 Jul 2025
21141	Hioki IM3533 LCR Meter	31 Oct 2024	31 Oct 2025
13714	Krohn Hite 6900B Distortion Analyzer	28 Feb 2025	28 Feb 2026
18432	Omega TJ36-ICIN-18U-12-SB-SMPW-M "J" Type Rugged Transition Joint Probe	05 Dec 2024	05 Dec 2025
18194	Tektronix FCA3103 Timer/Counter	04 Apr 2025	04 Apr 2026



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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	

INSTRUMENT INFORMATION:
FLUKE 5522A Serial Number 2806902
Firmware Revision: 1.3+1.3+1.8
Installed Options: SC600, 5520A-PQ

Test Definition Revision: 20240710

DC Voltage Accuracy Test (Normal Output)

330 mV Range

1 Year Specification

0.0000 mV	0.0000 mV	0.00013 mV	2.9E-007 V	-0.00100 mV	0.00100 mV
5.0000 mV	5.0000 mV	5.00010 mV	3.1E-007 V	4.99890 mV	5.00110 mV
20.0000 mV	20.0000 mV	19.99998 mV	4.1E-007 V	19.99860 mV	20.00140 mV
329.0000 mV	329.0000 mV	328.99940 mV	1.3E-006 V	328.99242 mV	329.00758 mV
-329.0000 mV	-329.0000 mV	-328.99780 mV	1.3E-006 V	-329.00758 mV	-328.99242 mV

3.3 V Range

1 Year Specification

0.000000 V	0.000000 V	-0.0000000 V	4E-007 V	-0.0000020 V	0.0000020 V
1.000000 V	1.000000 V	1.0000007 V	3.4E-006 V	0.9999870 V	1.0000130 V
-1.000000 V	-1.000000 V	-1.0000007 V	3.3E-006 V	-1.0000130 V	-0.9999870 V
3.290000 V	3.290000 V	3.2900019 V	1E-005 V	3.2899618 V	3.2900382 V
-3.290000 V	-3.290000 V	-3.2899983 V	1.1E-005 V	-3.2900382 V	-3.2899618 V

33 V Range

1 Year Specification

0.00000 V	0.00000 V	-0.0000009 V	3.2E-006 V	-0.000020 V	0.000020 V
10.00000 V	10.00000 V	9.999980 V	3.1E-005 V	9.999860 V	10.000140 V
-10.00000 V	-10.00000 V	-9.999995 V	3.1E-005 V	-10.000140 V	-9.999860 V
32.90000 V	32.90000 V	32.899834 V	4.7E-005 V	32.899585 V	32.900415 V
-32.90000 V	-32.90000 V	-32.899829 V	4.8E-005 V	-32.900415 V	-32.899585 V

330 V Range

1 Year Specification

50.0000 V	50.0000 V	50.00004 V	0.00025 V	49.99895 V	50.00105 V
-50.0000 V	-50.0000 V	-50.00015 V	0.00025 V	-50.00105 V	-49.99895 V
329.0000 V	329.0000 V	328.99949 V	0.00078 V	328.99393 V	329.00607 V
-329.0000 V	-329.0000 V	-328.99921 V	0.00078 V	-329.00607 V	-328.99393 V

1000 V Range

1 Year Specification

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
334.000 V	334.000 V	333.9998 V	0.0021 V	333.9925 V	334.0075 V	
-334.000 V	-334.000 V	-334.0009 V	0.0021 V	-334.0075 V	-333.9925 V	
900.000 V	900.000 V	899.9997 V	0.0046 V	899.9823 V	900.0177 V	
-900.000 V	-900.000 V	-900.0011 V	0.0046 V	-900.0177 V	-899.9823 V	
1000.000 V	1000.000 V	999.9994 V	0.0051 V	999.9805 V	1000.0195 V	
-1000.000 V	-1000.000 V	-1000.0003 V	0.0051 V	-1000.0195 V	-999.9805 V	

DC Voltage Accuracy Test (Aux Output)

330 mV Range

1 Year Specification

0.000 mV	0.000 mV	-0.0008 mV	2.9E-005 V	-0.3500 mV	0.3500 mV
329.000 mV	329.000 mV	328.9990 mV	1.3E-006 V	328.5184 mV	329.4816 mV
-329.000 mV	-329.000 mV	-329.0002 mV	1.4E-006 V	-329.4816 mV	-328.5184 mV

3.3 V Range

1 Year Specification

0.33000 V	0.33000 V	0.329999 V	2E-006 V	0.329518 V	0.330482 V
0.33000 V	-0.33000 V	-0.330005 V	1.6E-006 V	-0.330482 V	-0.329518 V
3.29000 V	3.29000 V	3.290024 V	1.2E-005 V	3.288334 V	3.291666 V
-3.29000 V	-3.29000 V	-3.290026 V	1.2E-005 V	-3.291666 V	-3.288334 V

7 V Range

1 Year Specification

7.0000 V	7.0000 V	7.00003 V	2.4E-005 V	6.99685 V	7.00315 V
-7.0000 V	-7.0000 V	-7.00004 V	2.5E-005 V	-7.00315 V	-6.99685 V

DC Voltage Offset

10 mV 1 kHz Signal

1 Year Specification

0.000 mV	0.000 mV	-0.0017 mV	3.1E-006 V	-0.0330 mV	0.0330 mV
50.000 mV	50.000 mV	49.9585 mV	3.1E-006 V	49.9170 mV	50.0830 mV
-50.000 mV	-50.000 mV	-49.9717 mV	3E-006 V	-50.0830 mV	-49.9170 mV

100 mV 1 kHz Signal

1 Year Specification

0.000 mV	0.000 mV	0.0050 mV	2.9E-005 V	-0.3300 mV	0.3300 mV
500.000 mV	500.000 mV	499.9681 mV	2.9E-005 V	499.1700 mV	500.8300 mV
-500.000 mV	-500.000 mV	-500.0464 mV	3.3E-005 V	-500.8300 mV	-499.1700 mV

1 V 1 kHz Signal

1 Year Specification

0.00000 V	0.00000 V	0.000064 V	0.00016 V	-0.003300 V	0.003300 V
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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
5.00000 V	5.00000 V	4.999913 V	0.00031 V	4.991700 V	5.008300 V	
-5.00000 V	-5.00000 V	-5.000647 V	0.0003 V	-5.008300 V	-4.991700 V	
3.3 V 1 kHz Signal						
1 Year Specification						
0.0000 V	0.0000 V	0.01546 V	0.0058 V	-0.03300 V	0.03300 V	
45.0000 V	45.0000 V	45.01826 V	0.0059 V	44.92200 V	45.07800 V	
-45.0000 V	-45.0000 V	-44.97596 V	0.0058 V	-45.07800 V	-44.92200 V	
50.0000 V	50.0000 V	50.01198 V	0.0059 V	49.91700 V	50.08300 V	
-50.0000 V	-50.0000 V	-49.96996 V	0.0059 V	-50.08300 V	-49.91700 V	
AC Voltage with DC Offset						
50 mV Offset						
1 Year Specification						
3.300 mV 1 kHz	3.300 mV	3.3015 mV	2.2E-006 V	3.2935 mV	3.3065 mV	
500 mV Offset						
1 Year Specification						
33.000 mV 1 kHz	33.000 mV	33.0036 mV	5.3E-006 V	32.9872 mV	33.0128 mV	
5 V Offset						
1 Year Specification						
0.3300 V 1 kHz	0.3300 V	0.33004 V	4.2E-005 V	0.32989 V	0.33011 V	
45 V Offset						
1 Year Specification						
3.3000 V 1 kHz	3.3000 V	3.29991 V	0.00042 V	3.29891 V	3.30109 V	
AC Voltage Accuracy Test (Normal Output)						
33 mV Range						
1 Year Specification						
3.000 mV 45 Hz	3.000 mV	3.0004 mV	1.5E-006 V	2.9935 mV	3.0065 mV	
3.000 mV 10 kHz	3.000 mV	3.0005 mV	1.5E-006 V	2.9935 mV	3.0065 mV	
30.000 mV 9.5 Hz	30.000 mV	30.0057 mV	2.5E-005 V	28.3350 mV	31.6650 mV	
30.000 mV 10 Hz	30.000 mV	29.9960 mV	6.8E-006 V	29.9700 mV	30.0300 mV	
30.000 mV 45 Hz	30.000 mV	30.0002 mV	2.7E-006 V	29.9895 mV	30.0105 mV	
30.000 mV 1 kHz	30.000 mV	30.0000 mV	2.7E-006 V	29.9895 mV	30.0105 mV	
30.000 mV 10 kHz	30.000 mV	30.0006 mV	2.7E-006 V	29.9895 mV	30.0105 mV	
30.000 mV 20 kHz	30.000 mV	30.0008 mV	2.8E-006 V	29.9880 mV	30.0120 mV	
30.000 mV 50 kHz	30.000 mV	29.9967 mV	4.7E-006 V	29.9640 mV	30.0360 mV	
30.000 mV 100 kHz	30.000 mV	29.9846 mV	8.2E-006 V	29.8830 mV	30.1170 mV	
30.000 mV 450 kHz	30.000 mV	29.9488 mV	4.6E-005 V	29.7100 mV	30.2900 mV	

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
30.000 mV 500 kHz	30.000 mV	30.0026 mV	2.2E-005 V	29.7100 mV	30.2900 mV	
330 mV Range						
1 Year Specification						
33.000 mV 45 Hz	33.000 mV	33.0001 mV	3.7E-006 V	32.9872 mV	33.0128 mV	
33.000 mV 10 kHz	33.000 mV	32.9998 mV	4.7E-006 V	32.9872 mV	33.0128 mV	
300.000 mV 9.5 Hz	300.000 mV	299.9962 mV	0.00024 V	283.3500 mV	316.6500 mV	
300.000 mV 10 Hz	300.000 mV	299.9644 mV	5E-005 V	299.9020 mV	300.0980 mV	
300.000 mV 45 Hz	300.000 mV	300.0006 mV	9.5E-006 V	299.9485 mV	300.0515 mV	
300.000 mV 1 kHz	300.000 mV	300.0026 mV	9.2E-006 V	299.9485 mV	300.0515 mV	
300.000 mV 10 kHz	300.000 mV	299.9980 mV	1.1E-005 V	299.9485 mV	300.0515 mV	
300.000 mV 20 kHz	300.000 mV	299.9924 mV	1.2E-005 V	299.9440 mV	300.0560 mV	
300.000 mV 50 kHz	300.000 mV	299.9718 mV	1.6E-005 V	299.8870 mV	300.1130 mV	
300.000 mV 100 kHz	300.000 mV	299.9510 mV	2.1E-005 V	299.7280 mV	300.2720 mV	
300.000 mV 500 kHz	300.000 mV	300.1546 mV	7.7E-005 V	299.3300 mV	300.6700 mV	
3.3 V Range						
1 Year Specification						
0.33000 V 45 Hz	0.33000 V	0.330020 V	2E-005 V	0.329891 V	0.330110 V	
0.33000 V 10 kHz	0.33000 V	0.330024 V	2.2E-005 V	0.329891 V	0.330110 V	
3.00000 V 9.5 Hz	3.00000 V	2.999764 V	0.0024 V	2.833500 V	3.166500 V	
3.00000 V 10 Hz	3.00000 V	2.999688 V	0.00047 V	2.999050 V	3.000950 V	
3.00000 V 45 Hz	3.00000 V	3.000016 V	6.4E-005 V	2.999490 V	3.000510 V	
3.00000 V 1 kHz	3.00000 V	3.000036 V	8.1E-005 V	2.999490 V	3.000510 V	
3.00000 V 10 kHz	3.00000 V	3.000060 V	6.6E-005 V	2.999490 V	3.000510 V	
3.00000 V 20 kHz	3.00000 V	3.000100 V	7.6E-005 V	2.999370 V	3.000630 V	
3.00000 V 50 kHz	3.00000 V	3.000280 V	0.00012 V	2.999050 V	3.000950 V	
3.00000 V 100 kHz	3.00000 V	3.000890 V	0.0002 V	2.997775 V	3.002225 V	
3.00000 V 450 kHz	3.00000 V	3.000876 V	0.00093 V	2.992200 V	3.007800 V	
3.00000 V 500 kHz	3.00000 V	3.000816 V	0.00093 V	2.992200 V	3.007800 V	
33 V Range						
1 Year Specification						
3.3000 V 45 Hz	3.3000 V	3.30020 V	0.00022 V	3.29891 V	3.30109 V	
3.3000 V 10 kHz	3.3000 V	3.30023 V	0.00018 V	3.29891 V	3.30109 V	
30.0000 V 9.5 Hz	30.0000 V	29.96916 V	0.025 V	28.33500 V	31.66500 V	
30.0000 V 10 Hz	30.0000 V	29.99644 V	0.0047 V	29.99035 V	30.00965 V	
30.0000 V 45 Hz	30.0000 V	30.00016 V	0.00075 V	29.99490 V	30.00510 V	
30.0000 V 1 kHz	30.0000 V	29.99862 V	0.0008 V	29.99490 V	30.00510 V	
30.0000 V 10 kHz	30.0000 V	29.99872 V	0.00083 V	29.99490 V	30.00510 V	
30.0000 V 20 kHz	30.0000 V	29.99908 V	0.00081 V	29.99220 V	30.00780 V	

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
30.0000 V 50 kHz	30.0000 V	29.99914 V	0.0015 V	29.98890 V	30.01110 V	
30.0000 V 100 kHz	30.0000 V	30.00012 V	0.0027 V	29.97140 V	30.02860 V	
330 V Range						
1 Year Specification						
33.000 V 45 Hz	33.000 V	33.0017 V	0.003 V	32.9917 V	33.0083 V	
33.000 V 10 kHz	33.000 V	33.0017 V	0.003 V	32.9874 V	33.0126 V	
300.000 V 45 Hz	300.000 V	300.0132 V	0.011 V	299.9410 V	300.0590 V	
300.000 V 1 kHz	300.000 V	299.9882 V	0.011 V	299.9410 V	300.0590 V	
300.000 V 10 kHz	300.000 V	299.9868 V	0.01 V	299.9340 V	300.0660 V	
300.000 V 17 kHz	300.000 V	299.9760 V	0.01 V	299.9190 V	300.0810 V	
300.000 V 18 kHz	300.000 V	299.9750 V	0.011 V	299.9190 V	300.0810 V	
300.000 V 20 kHz	300.000 V	299.9632 V	0.011 V	299.9190 V	300.0810 V	
300.000 V 50 kHz	300.000 V	299.9788 V	0.031 V	299.9040 V	300.0960 V	
200.000 V 100 kHz	200.000 V	200.0586 V	0.073 V	199.5500 V	200.4500 V	
1020 V Range						
1 Year Specification						
330.00 V 45 Hz	330.00 V	330.015 V	0.011 V	329.891 V	330.109 V	
330.00 V 10 kHz	330.00 V	330.006 V	0.023 V	329.891 V	330.109 V	
1000.00 V 45 Hz	1000.00 V	1000.167 V	0.03 V	999.690 V	1000.310 V	
1000.00 V 1 kHz	1000.00 V	1000.010 V	0.031 V	999.740 V	1000.260 V	
1000.00 V 5 kHz	1000.00 V	1000.002 V	0.031 V	999.740 V	1000.260 V	
1000.00 V 8 kHz	1000.00 V	999.988 V	0.033 V	999.690 V	1000.310 V	
1000.00 V 10 kHz	1000.00 V	999.983 V	0.055 V	999.690 V	1000.310 V	
1020.00 V 1 kHz	1020.00 V	1020.008 V	0.031 V	1019.735 V	1020.265 V	
1020.00 V 8 kHz	1020.00 V	1019.982 V	0.031 V	1019.684 V	1020.316 V	

AC Voltage Accuracy Test, Square Wave (Normal Output)

66 mV Range

1 Year Specification

60.000 mV 10 Hz	60.000 mVpp	60.0097 mVpp	4.9E-005 Vpp	59.5200 mVpp	60.4800 mVpp
60.000 mV 45 Hz	60.000 mVpp	60.0030 mVpp	4.9E-005 Vpp	59.6850 mVpp	60.3150 mVpp
60.000 mV 1 kHz	60.000 mVpp	60.0087 mVpp	4.9E-005 Vpp	59.6850 mVpp	60.3150 mVpp
60.000 mV 20 kHz	60.000 mVpp	60.1387 mVpp	8.2E-005 Vpp	59.5350 mVpp	60.4650 mVpp
60.000 mV 100 kHz	60.000 mVpp	60.7132 mVpp	0.00024 Vpp	56.6700 mVpp	63.3300 mVpp

660 mV Range

1 Year Specification

600.000 mV 10 Hz	600.000 mVpp	600.0707 mVpp	0.00047 Vpp	595.2000 mVpp	604.8000 mVpp
600.000 mV 45 Hz	600.000 mVpp	600.0227 mVpp	0.00047 Vpp	596.8500 mVpp	603.1500 mVpp

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
600.000 mV 1 kHz	600.000 mVpp	600.0927 mVpp	0.00047 Vpp	596.8500 mVpp	603.1500 mVpp	
600.000 mV 20 kHz	600.000 mVpp	601.5293 mVpp	0.00079 Vpp	595.3500 mVpp	604.6500 mVpp	
600.000 mV 100 kHz	600.000 mVpp	608.0267 mVpp	0.0023 Vpp	566.7000 mVpp	633.3000 mVpp	

6.6 V Range

1 Year Specification

6.00000 V 10 Hz	6.00000 Vpp	6.000800 Vpp	0.0047 Vpp	5.952000 Vpp	6.048000 Vpp	
6.00000 V 45 Hz	6.00000 Vpp	6.000280 Vpp	0.0047 Vpp	5.968500 Vpp	6.031500 Vpp	
6.00000 V 1 kHz	6.00000 Vpp	6.001180 Vpp	0.0047 Vpp	5.968500 Vpp	6.031500 Vpp	
6.00000 V 20 kHz	6.00000 Vpp	6.018840 Vpp	0.0079 Vpp	5.953500 Vpp	6.046500 Vpp	
6.00000 V 100 kHz	6.00000 Vpp	6.088053 Vpp	0.023 Vpp	5.667000 Vpp	6.333000 Vpp	

66 V Range

1 Year Specification

60.0000 V 10 Hz	60.0000 Vpp	60.00827 Vpp	0.047 Vpp	59.52000 Vpp	60.48000 Vpp	
60.0000 V 45 Hz	60.0000 Vpp	60.00267 Vpp	0.047 Vpp	59.68500 Vpp	60.31500 Vpp	
60.0000 V 1 kHz	60.0000 Vpp	60.00960 Vpp	0.047 Vpp	59.68500 Vpp	60.31500 Vpp	
60.0000 V 20 kHz	60.0000 Vpp	60.21767 Vpp	0.079 Vpp	59.53500 Vpp	60.46500 Vpp	
60.0000 V 100 kHz	60.0000 Vpp	61.13040 Vpp	0.23 Vpp	56.67000 Vpp	63.33000 Vpp	

AC Voltage Accuracy Test, Triangle Wave (Normal Output)

93 mV Range

1 Year Specification

90.000 mV 10 Hz	90.000 mVpp	89.9857 mVpp	7.4E-005 Vpp	89.3100 mVpp	90.6900 mVpp	
90.000 mV 45 Hz	90.000 mVpp	90.0015 mVpp	7.4E-005 Vpp	89.5425 mVpp	90.4575 mVpp	
90.000 mV 1 kHz	90.000 mVpp	90.0015 mVpp	7.4E-005 Vpp	89.5425 mVpp	90.4575 mVpp	
90.000 mV 20 kHz	90.000 mVpp	90.0076 mVpp	0.00012 Vpp	89.3175 mVpp	90.6825 mVpp	
90.000 mV 100 kHz	90.000 mVpp	90.0119 mVpp	0.00036 Vpp	85.0350 mVpp	94.9650 mVpp	

930 mV Range

1 Year Specification

900.000 mV 10 Hz	900.000 mVpp	899.8524 mVpp	0.00071 Vpp	893.1000 mVpp	906.9000 mVpp	
900.000 mV 45 Hz	900.000 mVpp	900.0025 mVpp	0.00071 Vpp	895.4250 mVpp	904.5750 mVpp	
900.000 mV 1 kHz	900.000 mVpp	900.0198 mVpp	0.00071 Vpp	895.4250 mVpp	904.5750 mVpp	
900.000 mV 20 kHz	900.000 mVpp	900.0336 mVpp	0.0012 Vpp	893.1750 mVpp	906.8250 mVpp	
900.000 mV 100 kHz	900.000 mVpp	900.4482 mVpp	0.0035 Vpp	850.3500 mVpp	949.6500 mVpp	

9.3 V Range

1 Year Specification

9.00000 V 10 Hz	9.00000 Vpp	8.998674 Vpp	0.007 Vpp	8.931000 Vpp	9.069000 Vpp	
9.00000 V 45 Hz	9.00000 Vpp	9.000059 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp	
9.00000 V 1 kHz	9.00000 Vpp	9.000279 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp	

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
9.00000 V 20 kHz	9.00000 Vpp	9.001087 Vpp	0.012 Vpp	8.931750 Vpp	9.068250 Vpp	
9.00000 V 100 kHz	9.00000 Vpp	9.011653 Vpp	0.035 Vpp	8.503500 Vpp	9.496500 Vpp	
93 V Range						
1 Year Specification						
90.0000 V 10 Hz	90.0000 Vpp	89.98628 Vpp	0.07 Vpp	89.31000 Vpp	90.69000 Vpp	
90.0000 V 45 Hz	90.0000 Vpp	90.00059 Vpp	0.07 Vpp	89.54250 Vpp	90.45750 Vpp	
90.0000 V 1 kHz	90.0000 Vpp	89.99736 Vpp	0.07 Vpp	89.54250 Vpp	90.45750 Vpp	
90.0000 V 20 kHz	90.0000 Vpp	90.01526 Vpp	0.12 Vpp	89.31750 Vpp	90.68250 Vpp	
90.0000 V 100 kHz	90.0000 Vpp	90.11537 Vpp	0.35 Vpp	85.03500 Vpp	94.96500 Vpp	

AC Voltage Accuracy Test, Truncated Wave (Normal Output)

93 mV Range

1 Year Specification

90.000 mV 10 Hz	90.000 mVpp	89.9385 mVpp	7.5E-005 Vpp	89.3100 mVpp	90.6900 mVpp
90.000 mV 45 Hz	90.000 mVpp	90.0000 mVpp	7.5E-005 Vpp	89.5425 mVpp	90.4575 mVpp
90.000 mV 1 kHz	90.000 mVpp	89.9963 mVpp	7.5E-005 Vpp	89.5425 mVpp	90.4575 mVpp
90.000 mV 20 kHz	90.000 mVpp	89.7658 mVpp	0.00012 Vpp	89.3175 mVpp	90.6825 mVpp

930 mV Range

1 Year Specification

900.000 mV 10 Hz	900.000 mVpp	899.5878 mVpp	0.00071 Vpp	893.1000 mVpp	906.9000 mVpp
900.000 mV 45 Hz	900.000 mVpp	899.9885 mVpp	0.00071 Vpp	895.4250 mVpp	904.5750 mVpp
900.000 mV 1 kHz	900.000 mVpp	899.9140 mVpp	0.00071 Vpp	895.4250 mVpp	904.5750 mVpp
900.000 mV 20 kHz	900.000 mVpp	898.0683 mVpp	0.0012 Vpp	893.1750 mVpp	906.8250 mVpp

9.3 V Range

1 Year Specification

9.00000 V 10 Hz	9.00000 Vpp	8.996657 Vpp	0.0097 Vpp	8.931000 Vpp	9.069000 Vpp
9.00000 V 45 Hz	9.00000 Vpp	8.998575 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp
9.00000 V 1 kHz	9.00000 Vpp	8.999807 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp
9.00000 V 20 kHz	9.00000 Vpp	8.982460 Vpp	0.012 Vpp	8.931750 Vpp	9.068250 Vpp

93 V Range

1 Year Specification

90.0000 V 10 Hz	90.0000 Vpp	89.93423 Vpp	0.095 Vpp	89.31000 Vpp	90.69000 Vpp
90.0000 V 45 Hz	90.0000 Vpp	89.91670 Vpp	0.07 Vpp	89.54250 Vpp	90.45750 Vpp
90.0000 V 1 kHz	90.0000 Vpp	89.93146 Vpp	0.07 Vpp	89.54250 Vpp	90.45750 Vpp
90.0000 V 20 kHz	90.0000 Vpp	89.82548 Vpp	0.12 Vpp	89.31750 Vpp	90.68250 Vpp

Harmonic Amplitude Accuracy Test (Normal Output)

33 mV Range

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
1 Year Specification						
30.000 mV 1 kHz	30.000 mV	30.0004 mV	2.7E-006 V	29.9835 mV	30.0165 mV	
30.000 mV 5 kHz	30.000 mV	30.0006 mV	2.7E-006 V	29.9835 mV	30.0165 mV	
30.000 mV 10 kHz	30.000 mV	30.0007 mV	2.7E-006 V	29.9835 mV	30.0165 mV	
30.000 mV 30 kHz	30.000 mV	29.9997 mV	4.6E-006 V	29.9580 mV	30.0420 mV	
330 mV Range						
1 Year Specification						
300.000 mV 1 kHz	300.000 mV	300.0043 mV	8.9E-006 V	299.9405 mV	300.0595 mV	
300.000 mV 5 kHz	300.000 mV	300.0027 mV	8.9E-006 V	299.9405 mV	300.0595 mV	
300.000 mV 10 kHz	300.000 mV	299.9983 mV	8.9E-006 V	299.9405 mV	300.0595 mV	
300.000 mV 30 kHz	300.000 mV	299.9833 mV	1.3E-005 V	299.8790 mV	300.1210 mV	
3.3 V Range						
1 Year Specification						
3.00000 V 1 kHz	3.00000 V	3.000057 V	5.8E-005 V	2.999430 V	3.000570 V	
3.00000 V 5 kHz	3.00000 V	3.000070 V	5.6E-005 V	2.999430 V	3.000570 V	
3.00000 V 10 kHz	3.00000 V	3.000070 V	5.6E-005 V	2.999430 V	3.000570 V	
3.00000 V 30 kHz	3.00000 V	3.000157 V	0.00011 V	2.999000 V	3.001000 V	
33 V Range						
1 Year Specification						
30.0000 V 1 kHz	30.0000 V	29.99890 V	0.00078 V	29.99430 V	30.00570 V	
30.0000 V 5 kHz	30.0000 V	29.99917 V	0.00075 V	29.99430 V	30.00570 V	
30.0000 V 10 kHz	30.0000 V	29.99887 V	0.00075 V	29.99430 V	30.00570 V	
30.0000 V 30 kHz	30.0000 V	29.99917 V	0.0013 V	29.98830 V	30.01170 V	
330 V Range						
1 Year Specification						
300.000 V 2.25 kHz	300.000 V	299.9913 V	0.0096 V	299.9280 V	300.0720 V	
300.000 V 5 kHz	300.000 V	299.9890 V	0.0096 V	299.9280 V	300.0720 V	
300.000 V 10 kHz	300.000 V	299.9883 V	0.0096 V	299.9280 V	300.0720 V	
300.000 V 30 kHz	300.000 V	299.9777 V	0.03 V	299.8980 V	300.1020 V	
1000 V Range						
1 Year Specification						
1000.00 V 2.25 kHz	1000.00 V	1000.038 V	0.03 V	999.730 V	1000.270 V	
1000.00 V 5 kHz	1000.00 V	1000.009 V	0.03 V	999.730 V	1000.270 V	
1000.00 V 10 kHz	1000.00 V	999.991 V	0.03 V	999.680 V	1000.320 V	

AC Voltage Accuracy Test (Aux Output)

330 mV Range

1 Year Specification

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
10.000 mV 45 Hz	10.000 mV	9.9994 mV	1E-005 V	9.6200 mV	10.3800 mV	
10.000 mV 1 kHz	10.000 mV	10.0000 mV	1E-005 V	9.6200 mV	10.3800 mV	
10.000 mV 5 kHz	10.000 mV	10.0033 mV	1E-005 V	9.5300 mV	10.4700 mV	
10.000 mV 10 kHz	10.000 mV	10.0050 mV	1E-005 V	9.5100 mV	10.4900 mV	
10.000 mV 30 kHz	10.000 mV	10.1450 mV	1.1E-005 V	8.6000 mV	11.4000 mV	
300.000 mV 9.5 Hz	300.000 mV	299.9827 mV	0.00024 V	283.3500 mV	316.6500 mV	
300.000 mV 10 Hz	300.000 mV	299.9197 mV	5.7E-005 V	299.0300 mV	300.9700 mV	
300.000 mV 20 Hz	300.000 mV	299.9913 mV	3.4E-005 V	299.3300 mV	300.6700 mV	
300.000 mV 45 Hz	300.000 mV	299.9953 mV	2.9E-005 V	299.3300 mV	300.6700 mV	
300.000 mV 1 kHz	300.000 mV	300.0010 mV	2.9E-005 V	299.3300 mV	300.6700 mV	
300.000 mV 5 kHz	300.000 mV	299.9980 mV	2.9E-005 V	298.9500 mV	301.0500 mV	
300.000 mV 10 kHz	300.000 mV	299.9957 mV	2.9E-005 V	298.3500 mV	301.6500 mV	
300.000 mV 30 kHz	300.000 mV	304.1237 mV	3.1E-005 V	284.1000 mV	315.9000 mV	

3.3 V Range

1 Year Specification

3.00000 V 9.5 Hz	3.00000 V	2.999690 V	0.0023 V	2.825000 V	3.175000 V
3.00000 V 10 Hz	3.00000 V	2.999183 V	0.00055 V	2.993550 V	3.006450 V
3.00000 V 20 Hz	3.00000 V	2.999917 V	0.00032 V	2.996550 V	3.003450 V
3.00000 V 45 Hz	3.00000 V	2.999937 V	0.00029 V	2.996850 V	3.003150 V
3.00000 V 1 kHz	3.00000 V	2.999957 V	0.00029 V	2.996850 V	3.003150 V
3.00000 V 5 kHz	3.00000 V	2.999917 V	0.00029 V	2.992600 V	3.007400 V
3.00000 V 10 kHz	3.00000 V	2.999890 V	0.00029 V	2.986600 V	3.013400 V
3.00000 V 30 kHz	3.00000 V	3.001410 V	0.00031 V	2.847200 V	3.152800 V

5 V Range

1 Year Specification

5.0000 V 9.5 Hz	5.0000 V	4.99703 V	0.0039 V	4.72500 V	5.27500 V
5.0000 V 10 Hz	5.0000 V	4.99875 V	0.00089 V	4.98955 V	5.01045 V
5.0000 V 20 Hz	5.0000 V	4.99989 V	0.0005 V	4.99455 V	5.00545 V
5.0000 V 45 Hz	5.0000 V	4.99995 V	0.00041 V	4.99505 V	5.00495 V
5.0000 V 1 kHz	5.0000 V	5.00033 V	0.00041 V	4.99505 V	5.00495 V
5.0000 V 5 kHz	5.0000 V	5.00009 V	0.00041 V	4.98860 V	5.01140 V
5.0000 V 10 kHz	5.0000 V	5.00027 V	0.00041 V	4.97860 V	5.02140 V

AC Voltage Accuracy Test, Square Wave (Aux Output)

660 mV Range

1 Year Specification

600.000 mV 10 Hz	600.000 mVpp	600.0673 mVpp	0.00047 Vpp	595.2000 mVpp	604.8000 mVpp
600.000 mV 45 Hz	600.000 mVpp	600.0547 mVpp	0.00047 Vpp	596.8500 mVpp	603.1500 mVpp

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
600.000 mV 1 kHz	600.000 mVpp	601.0780 mVpp	0.00079 Vpp	596.8500 mVpp	603.1500 mVpp	
600.000 mV 10 kHz	600.000 mVpp	611.7413 mVpp	0.00079 Vpp	566.7000 mVpp	633.3000 mVpp	

6.6 V Range

1 Year Specification

6.00000 V 10 Hz	6.00000 Vpp	6.000740 Vpp	0.0047 Vpp	5.952000 Vpp	6.048000 Vpp	
6.00000 V 45 Hz	6.00000 Vpp	6.000513 Vpp	0.0047 Vpp	5.968500 Vpp	6.031500 Vpp	
6.00000 V 1 kHz	6.00000 Vpp	6.009633 Vpp	0.0047 Vpp	5.968500 Vpp	6.031500 Vpp	
6.00000 V 10 kHz	6.00000 Vpp	6.097560 Vpp	0.0079 Vpp	5.667000 Vpp	6.333000 Vpp	

AC Voltage Accuracy Test, Triangle Wave (Aux Output)

930 mV Range

1 Year Specification

900.000 mV 10 Hz	900.000 mVpp	899.7046 mVpp	0.00071 Vpp	893.1000 mVpp	906.9000 mVpp	
900.000 mV 45 Hz	900.000 mVpp	899.9851 mVpp	0.00071 Vpp	895.4250 mVpp	904.5750 mVpp	
900.000 mV 1 kHz	900.000 mVpp	900.0406 mVpp	0.0012 Vpp	895.4250 mVpp	904.5750 mVpp	
900.000 mV 10 kHz	900.000 mVpp	900.0487 mVpp	0.0012 Vpp	850.3500 mVpp	949.6500 mVpp	

9.3 V Range

1 Year Specification

9.00000 V 10 Hz	9.00000 Vpp	8.997126 Vpp	0.007 Vpp	8.931000 Vpp	9.069000 Vpp	
9.00000 V 45 Hz	9.00000 Vpp	8.999921 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp	
9.00000 V 1 kHz	9.00000 Vpp	8.999921 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp	
9.00000 V 10 kHz	9.00000 Vpp	8.983201 Vpp	0.012 Vpp	8.503500 Vpp	9.496500 Vpp	

AC Voltage Accuracy Test, Truncated Wave (Aux Output)

930 mV Range

1 Year Specification

900.000 mV 10 Hz	900.000 mVpp	898.9442 mVpp	0.00072 Vpp	893.1000 mVpp	906.9000 mVpp	
900.000 mV 45 Hz	900.000 mVpp	899.4459 mVpp	0.00071 Vpp	895.4250 mVpp	904.5750 mVpp	
900.000 mV 1 kHz	900.000 mVpp	899.1273 mVpp	0.0012 Vpp	895.4250 mVpp	904.5750 mVpp	
900.000 mV 10 kHz	900.000 mVpp	870.4329 mVpp	0.0012 Vpp	850.3500 mVpp	949.6500 mVpp	

9.3 V Range

1 Year Specification

9.00000 V 10 Hz	9.00000 Vpp	8.988651 Vpp	0.0071 Vpp	8.931000 Vpp	9.069000 Vpp	
9.00000 V 45 Hz	9.00000 Vpp	8.994416 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp	
9.00000 V 1 kHz	9.00000 Vpp	8.990615 Vpp	0.007 Vpp	8.954250 Vpp	9.045750 Vpp	
9.00000 V 10 kHz	9.00000 Vpp	8.638630 Vpp	0.012 Vpp	8.503500 Vpp	9.496500 Vpp	

Harmonic Amplitude Accuracy Test (Aux Output)

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
330 mV Range						
1 Year Specification						
300.000 mV 1 kHz	300.000 mV	300.0027 mV	1.3E-005 V	298.9600 mV	301.0400 mV	
300.000 mV 5 kHz	300.000 mV	300.0003 mV	0.00012 V	298.5000 mV	301.5000 mV	
300.000 mV 10 kHz	300.000 mV	299.9993 mV	0.00017 V	297.9000 mV	302.1000 mV	
300.000 mV 30 kHz	300.000 mV	304.1437 mV	0.00017 V	283.2000 mV	316.8000 mV	
3.3 V Range						
1 Year Specification						
3.00000 V 1 kHz	3.00000 V	2.999983 V	0.00013 V	2.996400 V	3.003600 V	
3.00000 V 5 kHz	3.00000 V	2.999957 V	0.00024 V	2.991200 V	3.008800 V	
3.00000 V 10 kHz	3.00000 V	2.999940 V	0.00024 V	2.985200 V	3.014800 V	
3.00000 V 30 kHz	3.00000 V	3.001477 V	0.00017 V	2.844400 V	3.155600 V	
5 V Range						
1 Year Specification						
5.0000 V 1 kHz	5.0000 V	5.00062 V	0.00091 V	4.99460 V	5.00540 V	
5.0000 V 5 kHz	5.0000 V	4.99997 V	0.001 V	4.98720 V	5.01280 V	
5.0000 V 10 kHz	5.0000 V	4.99971 V	0.004 V	4.97720 V	5.02280 V	

Distortion Accuracy Test (Normal Output)

330 mV Range

1 Year Specification

300 mV 10 Hz < 0.180%: Measured = 0.048	Pass
300 mV 45 Hz < 0.065%: Measured = 0.010	Pass
300 mV 10 kHz < 0.065%: Measured = 0.008	Pass
300 mV 20 kHz < 0.090%: Measured = 0.013	Pass
300 mV 50 kHz < 0.180%: Measured = 0.030	Pass
300 mV 100 kHz < 0.230%: Measured = 0.046	Pass
300 mV 500 kHz < 0.930%: Measured = 0.112	Pass

3.3 V Range

1 Year Specification

1 V 10 Hz < 0.170%: Measured = 0.050	Pass
1 V 45 Hz < 0.055%: Measured = 0.010	Pass
1 V 10 kHz < 0.055%: Measured = 0.011	Pass
1 V 20 kHz < 0.080%: Measured = 0.010	Pass
1 V 50 kHz < 0.170%: Measured = 0.022	Pass
1 V 100 kHz < 0.220%: Measured = 0.056	Pass
1 V 500 kHz < 0.920%: Measured = 0.152	Pass

330 V Range

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Calibration Results

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				Lower Limit	Upper Limit	
1 Year Specification						
120 V 45 Hz < 0.158%: Measured = 0.008						Pass
120 V 1 kHz < 0.058%: Measured = 0.009						Pass
120 V 10 kHz < 0.058%: Measured = 0.025						Pass
120 V 20 kHz < 0.608%: Measured = 0.043						Pass
120 V 50 kHz < 0.808%: Measured = 0.059						Pass
120 V 100 kHz < 1.008%: Measured = 0.128						Pass

Distortion Accuracy Test (Aux Output)

330 mV Range

1 Year Specification						
300 mV 10 Hz < 0.267%: Measured = 0.107						Pass
300 mV 20 Hz < 0.127%: Measured = 0.015						Pass
300 mV 45 Hz < 0.127%: Measured = 0.009						Pass
300 mV 1 kHz < 0.147%: Measured = 0.007						Pass
300 mV 5 kHz < 0.367%: Measured = 0.023						Pass
300 mV 10 kHz < 0.667%: Measured = 0.042						Pass
300 mV 30 kHz < 1.067%: Measured = 0.067						Pass

3.3 V Range

1 Year Specification						
1 V 10 Hz < 0.220%: Measured = 0.107						Pass
1 V 20 Hz < 0.080%: Measured = 0.011						Pass
1 V 45 Hz < 0.080%: Measured = 0.006						Pass
1 V 1 kHz < 0.100%: Measured = 0.004						Pass
1 V 5 kHz < 0.320%: Measured = 0.006						Pass
1 V 10 kHz < 0.620%: Measured = 0.007						Pass
1 V 30 kHz < 1.020%: Measured = 0.061						Pass

5 V Range

1 Year Specification						
5 V 10 Hz < 0.204%: Measured = 0.107						Pass
5 V 20 Hz < 0.064%: Measured = 0.009						Pass
5 V 45 Hz < 0.064%: Measured = 0.007						Pass
5 V 1 kHz < 0.084%: Measured = 0.006						Pass
5 V 5 kHz < 0.304%: Measured = 0.156						Pass
5 V 10 kHz < 0.604%: Measured = 0.291						Pass

Frequency Accuracy Test

1 Year Specification

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Calibration Results

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				Lower Limit	Upper Limit	
2.000 MHz 3 V	2.000000 MHz	1.9999993 MHz	0.55 Hz	1.9999950 MHz	2.0000050 MHz	
500.0 kHz 3 V	500.00000 kHz	499.999813 kHz	0.14 Hz	499.998750 kHz	500.001250 kHz	
100.00 kHz 3 V	100.00000 kHz	99.999963 kHz	0.028 Hz	99.999750 kHz	100.000250 kHz	
1000.0 Hz 3 V	1000.0000 Hz	999.99962 Hz	0.00028 Hz	999.99750 Hz	1000.00250 Hz	
120.0 Hz 3 V	120.000000 Hz	119.9999558 Hz	3.5E-005 Hz	119.9996950 Hz	120.0003050 Hz	
119.00 Hz 3 V	119.000000 Hz	118.9999538 Hz	3.3E-005 Hz	118.9996975 Hz	119.0003025 Hz	
60.00 Hz 3 V	60.000000 Hz	59.9999871 Hz	1.8E-005 Hz	59.9998450 Hz	60.0001550 Hz	
50.00 Hz 3 V	50.000000 Hz	49.9999815 Hz	1.7E-005 Hz	49.9998700 Hz	50.0001300 Hz	

Capacitance Accuracy Test

400 pF Range

1 Year Specification

220.0 pF 5 KHz	220.0 pF	219.64 pF	1.5E-012 F	208.90 pF	231.10 pF
390.0 pF 1 KHz	390.0 pF	390.03 pF	1.6E-012 F	378.05 pF	401.95 pF

1.1 nF Range

1 Year Specification

0.4800 nF 1 KHz	0.4800 nF	0.48006 nF	1.9E-012 F	0.46760 nF	0.49240 nF
0.6000 nF 1 KHz	0.6000 nF	0.60002 nF	1E-012 F	0.58700 nF	0.61300 nF
1.0000 nF 1 KHz	1.0000 nF	0.99996 nF	5E-013 F	0.98500 nF	1.01500 nF

3.3 nF Range

1 Year Specification

2.0000 nF 1 KHz	2.0000 nF	1.99957 nF	1E-012 F	1.98000 nF	2.02000 nF
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11 nF Range

1 Year Specification

7.0000 nF 1 KHz	7.0000 nF	6.99949 nF	3.5E-012 F	6.97250 nF	7.02750 nF
10.9000 nF 1 KHz	10.9000 nF	10.89930 nF	5.4E-012 F	10.86275 nF	10.93725 nF

33 nF Range

1 Year Specification

20.0000 nF 1 KHz	20.0000 nF	19.99693 nF	1E-011 F	19.85000 nF	20.15000 nF
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110 nF Range

1 Year Specification

70.000 nF 1 KHz	70.000 nF	69.9948 nF	3.5E-011 F	69.7250 nF	70.2750 nF
109.000 nF 1 KHz	109.000 nF	108.9940 nF	5.5E-011 F	108.6275 nF	109.3725 nF

330 nF Range

1 Year Specification

200.000 nF 1 KHz	200.000 nF	199.9490 nF	1E-010 F	199.2000 nF	200.8000 nF
300.000 nF 1 KHz	300.000 nF	299.9720 nF	1.5E-010 F	298.9500 nF	301.0500 nF

1.1 µF Range

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Calibration Results

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Lower LimitUpper Limit						
1 Year Specification						
0.70000 uF 100 Hz	0.70000 µF	0.700003 µF	5.6E-010 F	0.697250 µF	0.702750 µF	
1.09000 uF 100 Hz	1.09000 µF	1.090000 µF	6.5E-010 F	1.086275 µF	1.093725 µF	
3.3 µF Range						
1 Year Specification						
2.00000 uF 100 Hz	2.00000 µF	2.000077 µF	1.2E-009 F	1.992000 µF	2.008000 µF	
3.00000 uF 100 Hz	3.00000 µF	3.000280 µF	1.8E-009 F	2.989500 µF	3.010500 µF	
11 µF Range						
1 Year Specification						
7.0000 uF 100 Hz	7.0000 µF	7.00100 µF	4.2E-009 F	6.97250 µF	7.02750 µF	
10.9000 uF 100 Hz	10.9000 µF	10.90310 µF	8.7E-009 F	10.86275 µF	10.93725 µF	
33 µF Range						
1 Year Specification						
20.0000 uF 100 Hz	20.0000 µF	20.00780 µF	1.6E-008 F	19.89000 µF	20.11000 µF	
30.0000 uF 100 Hz	30.0000 µF	30.02340 µF	2.4E-008 F	29.85000 µF	30.15000 µF	
110 µF Range						
1 Year Specification						
70.000 uF 50 Hz	70.000 µF	69.9691 µF	5.6E-008 F	69.5850 µF	70.4150 µF	
109.000 uF 50 Hz	109.000 µF	108.9920 µF	1.2E-007 F	108.4095 µF	109.5905 µF	
330 µF Range						
1 Year Specification						
200.000 uF 40 Hz	200.000 µF	200.0190 µF	2.2E-007 F	198.8000 µF	201.2000 µF	
300.000 uF 40 Hz	300.000 µF	300.2170 µF	3.3E-007 F	298.3500 µF	301.6500 µF	
1.1 mF Range						
1 Year Specification						
0.33000 mF 20 Hz	0.33000 mF	0.329759 mF	5.9E-007 F	0.327515 mF	0.332485 mF	
0.70000 mF 20 Hz	0.70000 mF	0.699914 mF	5.6E-007 F	0.695850 mF	0.704150 mF	
1.09000 mF 15 Hz	1.09000 mF	1.089710 mF	7.3E-007 F	1.084095 mF	1.095905 mF	
3.3 mF Range						
1 Year Specification						
1.10000 mF 6 Hz	1.10000 mF	1.098900 mF	7.4E-007 F	1.092050 mF	1.107950 mF	
2.00000 mF 6 Hz	2.00000 mF	1.998253 mF	1.2E-006 F	1.988000 mF	2.012000 mF	
3.00000 mF 6 Hz	3.00000 mF	2.997777 mF	1.5E-006 F	2.983500 mF	3.016500 mF	
11 mF Range						
1 Year Specification						
3.3000 mF 2 Hz	3.3000 mF	3.29666 mF	1.7E-006 F	3.27515 mF	3.32485 mF	
10.9000 mF 2 Hz	10.9000 mF	10.88990 mF	5.5E-006 F	10.84095 mF	10.95905 mF	
33 mF Range						



Certificate Number: EVL1056908

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
1 Year Specification						
20.0000 mF 0.6 Hz	20.0000 mF	19.98203 mF	1E-005 F	19.82000 mF	20.18000 mF	
30.0000 mF 0.6 Hz	30.0000 mF	29.97350 mF	1.5E-005 F	29.74500 mF	30.25500 mF	
110 mF Range						
1 Year Specification						
33.00 mF 0.2 Hz	33.00 mF	32.980 mF	1.7E-005 F	32.537 mF	33.463 mF	
110.00 mF 0.2 Hz	110.00 mF	109.928 mF	5.5E-005 F	108.690 mF	111.310 mF	

Thermocouple Source Accuracy Test

10 $\mu\text{V}/^\circ\text{C}$

1 Year Specification

0.0000 mV	0.0000 mV	-0.00006 mV	2.1E-007 V	-0.00300 mV	0.00300 mV
1.0000 mV	1.0000 mV	0.99993 mV	2.5E-007 V	0.99695 mV	1.00305 mV
-1.0000 mV	-1.0000 mV	-1.00012 mV	2.3E-007 V	-1.00305 mV	-0.99695 mV
10.0000 mV	10.0000 mV	9.99975 mV	3E-007 V	9.99650 mV	10.00350 mV
-10.0000 mV	-10.0000 mV	-9.99999 mV	3.4E-007 V	-10.00350 mV	-9.99650 mV
100.0000 mV	100.0000 mV	99.99971 mV	9.7E-007 V	99.99200 mV	100.00800 mV
-100.0000 mV	-100.0000 mV	-99.99979 mV	9.7E-007 V	-100.00800 mV	-99.99200 mV

Thermocouple Measure Accuracy Test

10 $\mu\text{V}/^\circ\text{C}$

1 Year Specification

0.0000 mV	0.00007 mV	0.0003 mV	4.3E-007 V	-0.0029 mV	0.0031 mV
1.0000 mV	1.00013 mV	1.0000 mV	4.4E-007 V	0.9971 mV	1.0032 mV
-1.0000 mV	-0.99956 mV	-0.9996 mV	2.9E-007 V	-1.0026 mV	-0.9965 mV
10.0000 mV	9.99989 mV	10.0000 mV	4E-007 V	9.9964 mV	10.0034 mV
-10.0000 mV	-9.99923 mV	-9.9993 mV	4.1E-007 V	-10.0027 mV	-9.9957 mV
100.0000 mV	99.99979 mV	100.0003 mV	9.9E-007 V	99.9918 mV	100.0078 mV
-100.0000 mV	-99.99821 mV	-99.9994 mV	9.9E-007 V	-100.0062 mV	-99.9902 mV

Thermocouple Cold Junction Compensation Accuracy Test

1 Year Specification

23.00° Ambient (°C)	23.132 °C	23.14 °C	0.038 °C	22.99 °C	23.27 °C
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Resistance Accuracy Test

11 Ohm Range

1 Year Specification

2-Wire Compensation Test

Pass

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Date of Calibration: 29 May 2025

Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
0.0000 Ohm	0.0000 Ω	-0.00002 Ω	6E-005 Ω	-0.00100 Ω	0.00100 Ω	
2.0000 Ohm	2.0000 Ω	1.99974 Ω	0.00041 Ω	1.99892 Ω	2.00108 Ω	
10.9000 Ohm	10.9000 Ω	10.90007 Ω	0.00037 Ω	10.89856 Ω	10.90144 Ω	
33 Ohm Range						
1 Year Specification						
11.9000 Ohm	11.9000 Ω	11.90007 Ω	0.00036 Ω	11.89814 Ω	11.90186 Ω	
19.0000 Ohm	19.0000 Ω	19.00007 Ω	0.00041 Ω	18.99793 Ω	19.00207 Ω	
30.0000 Ohm	30.0000 Ω	30.00005 Ω	0.00046 Ω	29.99760 Ω	30.00240 Ω	
110 Ohm Range						
1 Year Specification						
33.0000 Ohm	33.0000 Ω	33.00001 Ω	0.00039 Ω	32.99768 Ω	33.00232 Ω	
109.0000 Ohm	109.0000 Ω	109.00002 Ω	0.0011 Ω	108.99555 Ω	109.00445 Ω	
330 Ohm Range						
1 Year Specification						
119.0000 Ohm	119.0000 Ω	118.99965 Ω	0.0012 Ω	118.99467 Ω	119.00533 Ω	
190.0000 Ohm	190.0000 Ω	189.99891 Ω	0.0019 Ω	189.99268 Ω	190.00732 Ω	
300.0000 Ohm	300.0000 Ω	300.00022 Ω	0.0036 Ω	299.98960 Ω	300.01040 Ω	
1.1 kOhm Range						
1 Year Specification						
0.330000 kOhm	0.330000 k Ω	0.3299989 k Ω	0.0039 Ω	0.3299888 k Ω	0.3300112 k Ω	
1.090000 kOhm	1.090000 k Ω	1.0899990 k Ω	0.011 Ω	1.0899675 k Ω	1.0900325 k Ω	
3.3 kOhm Range						
1 Year Specification						
1.190000 kOhm	1.190000 k Ω	1.1899980 k Ω	0.012 Ω	1.1899467 k Ω	1.1900533 k Ω	
1.900000 kOhm	1.900000 k Ω	1.8999949 k Ω	0.018 Ω	1.8999268 k Ω	1.9000732 k Ω	
3.000000 kOhm	3.000000 k Ω	3.0000026 k Ω	0.037 Ω	2.9998960 k Ω	3.0001040 k Ω	
11 kOhm Range						
1 Year Specification						
3.30000 kOhm	3.30000 k Ω	3.299987 k Ω	0.039 Ω	3.299888 k Ω	3.300112 k Ω	
10.90000 kOhm	10.90000 k Ω	10.900000 k Ω	0.11 Ω	10.899675 k Ω	10.900325 k Ω	
33 kOhm Range						
1 Year Specification						
11.90000 kOhm	11.90000 k Ω	11.899986 k Ω	0.12 Ω	11.899467 k Ω	11.900533 k Ω	
19.00000 kOhm	19.00000 k Ω	18.999991 k Ω	0.19 Ω	18.999268 k Ω	19.000732 k Ω	
30.00000 kOhm	30.00000 k Ω	29.999984 k Ω	0.37 Ω	29.998960 k Ω	30.001040 k Ω	
110 kOhm Range						
1 Year Specification						
33.0000 kOhm	33.0000 k Ω	32.99996 k Ω	0.4 Ω	32.99888 k Ω	33.00112 k Ω	

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
Lower Limit	Upper Limit					
109.0000 kOhm	109.0000 kΩ	108.99998 kΩ	1.1 Ω	108.99675 kΩ	109.00325 kΩ	
330 kOhm Range						
1 Year Specification						
119.0000 kOhm	119.0000 kΩ	118.99930 kΩ	1.2 Ω	118.99419 kΩ	119.00581 kΩ	
190.0000 kOhm	190.0000 kΩ	189.99957 kΩ	2 Ω	189.99192 kΩ	190.00808 kΩ	
300.0000 kOhm	300.0000 kΩ	299.99974 kΩ	4.6 Ω	299.98840 kΩ	300.01160 kΩ	
1.1 MOhm Range						
1 Year Specification						
0.330000 MOhm	0.330000 MΩ	0.3300029 MΩ	5 Ω	0.3299874 MΩ	0.3300126 MΩ	
1.090000 MOhm	1.090000 MΩ	1.0899978 MΩ	13 Ω	1.0899631 MΩ	1.0900369 MΩ	
3.3 MOhm Range						
1 Year Specification						
1.190000 MOhm	1.190000 MΩ	1.1899866 MΩ	14 Ω	1.1898986 MΩ	1.1901014 MΩ	
1.900000 MOhm	1.900000 MΩ	1.8999823 MΩ	22 Ω	1.8998560 MΩ	1.9001440 MΩ	
3.000000 MOhm	3.000000 MΩ	2.9999896 MΩ	33 Ω	2.9997900 MΩ	3.0002100 MΩ	
11 MOhm Range						
1 Year Specification						
3.30000 MOhm	3.30000 MΩ	3.299984 MΩ	37 Ω	3.299521 MΩ	3.300479 MΩ	
10.9000 MOhm	10.90000 MΩ	10.899967 MΩ	3.1E+002 Ω	10.898533 MΩ	10.901467 MΩ	
33 MOhm Range						
1 Year Specification						
11.90000 MOhm	11.90000 MΩ	11.899812 MΩ	2E+003 Ω	11.894525 MΩ	11.905475 MΩ	
19.00000 MOhm	19.00000 MΩ	18.999587 MΩ	2.1E+003 Ω	18.992750 MΩ	19.007250 MΩ	
30.00000 MOhm	30.00000 MΩ	29.999264 MΩ	1.1E+003 Ω	29.990000 MΩ	30.010000 MΩ	
110 MOhm Range						
1 Year Specification						
33.0000 MOhm	33.0000 MΩ	32.99831 MΩ	2.1E+003 Ω	32.98050 MΩ	33.01950 MΩ	
109.0000 MOhm	109.0000 MΩ	108.99460 MΩ	2.3E+004 Ω	108.94250 MΩ	109.05750 MΩ	
330 MOhm Range						
1 Year Specification						
119.000 MOhm	119.000 MΩ	118.9109 MΩ	1.7E+005 Ω	118.5430 MΩ	119.4570 MΩ	
190.000 MOhm	190.000 MΩ	189.8803 MΩ	3.3E+004 Ω	189.3300 MΩ	190.6700 MΩ	
300.000 MOhm	300.000 MΩ	299.8285 MΩ	5.6E+004 Ω	299.0000 MΩ	301.0000 MΩ	
1100 MOhm Range						
1 Year Specification						
400.00 MOhm	400.00 MΩ	400.067 MΩ	1.5E+006 Ω	393.500 MΩ	406.500 MΩ	
640.00 MOhm	640.00 MΩ	639.945 MΩ	1.8E+006 Ω	629.900 MΩ	650.100 MΩ	
1090.00 MOhm	1090.00 MΩ	1089.675 MΩ	2.4E+006 Ω	1073.150 MΩ	1106.850 MΩ	

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
Phase Accuracy Test						
Volts Output Normal, Volts Output Aux						
1 Year Specification						
0.00° 3 V, 3 V, 65 Hz	0.00 °	-0.014 °	0.02 °	-0.100 °	0.100 °	
0.00° 3 V, 3 V, 400 Hz	0.00 °	-0.002 °	0.02 °	-0.250 °	0.250 °	
0.00° 3 V, 3 V, 1 kHz	0.00 °	-0.001 °	0.02 °	-0.500 °	0.500 °	
0.00° 3 V, 3 V, 5 kHz	0.00 °	0.072 °	0.03 °	-2.500 °	2.500 °	
0.00° 3 V, 3 V, 10 kHz	0.00 °	0.131 °	0.04 °	-5.000 °	5.000 °	
0.00° 3 V, 3 V, 30 kHz	0.00 °	0.677 °	0.051 °	-10.000 °	10.000 °	
60.00° 3 V, 3 V, 65 Hz	60.00 °	59.992 °	0.02 °	59.900 °	60.100 °	
60.00° 3 V, 3 V, 400 Hz	60.00 °	59.999 °	0.02 °	59.750 °	60.250 °	
60.00° 3 V, 3 V, 1 kHz	60.00 °	60.012 °	0.02 °	59.500 °	60.500 °	
60.00° 3 V, 3 V, 5 kHz	60.00 °	60.135 °	0.03 °	57.500 °	62.500 °	
60.00° 3 V, 3 V, 10 kHz	60.00 °	60.293 °	0.061 °	55.000 °	65.000 °	
60.00° 3 V, 3 V, 30 kHz	60.00 °	60.710 °	0.053 °	50.000 °	70.000 °	
90.00° 3 V, 3 V, 65 Hz	90.00 °	89.987 °	0.02 °	89.900 °	90.100 °	
90.00° 3 V, 3 V, 400 Hz	90.00 °	89.999 °	0.02 °	89.750 °	90.250 °	
90.00° 3 V, 3 V, 1 kHz	90.00 °	90.005 °	0.03 °	89.500 °	90.500 °	
90.00° 3 V, 3 V, 5 kHz	90.00 °	90.073 °	0.03 °	87.500 °	92.500 °	
90.00° 3 V, 3 V, 10 kHz	90.00 °	90.131 °	0.04 °	85.000 °	95.000 °	
90.00° 3 V, 3 V, 30 kHz	90.00 °	90.791 °	0.05 °	80.000 °	100.000 °	
90.00° 30 V, 3 V, 65 Hz	90.00 °	89.971 °	0.02 °	89.900 °	90.100 °	
90.00° 50 V, 3 V, 65 Hz	90.00 °	89.971 °	0.027 °	89.900 °	90.100 °	

Phase Accuracy Test

Volts Output Normal, Amps Output Aux

1 Year Specification

0.00° 30 mV, 300 mA, 65 Hz	0.00 °	-0.003 °	0.02 °	-0.100 °	0.100 °	
0.00° 30 mV, 300 mA, 1 kHz	0.00 °	0.067 °	0.024 °	-0.500 °	0.500 °	
0.00° 30 mV, 300 mA, 30 kHz	0.00 °	1.930 °	0.9 °	-10.000 °	10.000 °	
0.00° 200 mV, 2 A, 65 Hz	0.00 °	0.004 °	0.02 °	-0.100 °	0.100 °	
0.00° 33 V, 300 mA, 65 Hz	0.00 °	-0.022 °	0.025 °	-0.100 °	0.100 °	
0.00° 33 V, 2 A, 65 Hz	0.00 °	-0.019 °	0.027 °	-0.100 °	0.100 °	
0.00° 50 mV, 5 A, 65 Hz	0.00 °	-0.015 °	0.02 °	-0.100 °	0.100 °	
0.00° 50 mV, 5 A, 400 Hz	0.00 °	-0.112 °	0.034 °	-0.250 °	0.250 °	
0.00° 33 V, 5 A, 65 Hz	0.00 °	-0.032 °	0.021 °	-0.100 °	0.100 °	
0.00° 33 V, 5 A, 400 Hz	0.00 °	-0.093 °	0.02 °	-0.250 °	0.250 °	



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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
60.00° 30 mV, 300 mA, 65 Hz	60.00 °	59.999 °	0.02 °	59.900 °	60.100 °	
60.00° 200 mV, 2 A, 65 Hz	60.00 °	60.005 °	0.02 °	59.900 °	60.100 °	
60.00° 200 mV, 20 A, 65 Hz	60.00 °	59.994 °	0.02 °	59.900 °	60.100 °	
60.00° 200 mV, 20 A, 400 Hz	60.00 °	59.924 °	0.026 °	59.750 °	60.250 °	
90.00° 33 V, 300 mA, 65 Hz	90.00 °	89.975 °	0.02 °	89.900 °	90.100 °	
90.00° 33 V, 2 A, 65 Hz	90.00 °	89.980 °	0.03 °	89.900 °	90.100 °	
90.00° 33 V, 5 A, 400 Hz	90.00 °	89.905 °	0.02 °	89.750 °	90.250 °	
90.00° 3.3 V, 20 A, 65 Hz	90.00 °	89.977 °	0.02 °	89.900 °	90.100 °	
90.00° 33 V, 5 A, 65 Hz	90.00 °	89.971 °	0.04 °	89.900 °	90.100 °	
90.00° 33 V, 20 A, 400 Hz	90.00 °	89.895 °	0.025 °	89.750 °	90.250 °	

DC Current Accuracy Test

330 µA Range

1 Year Specification

0.000 µA	0.000 µA	-0.0013 µA	4.1E-010 A	-0.0200 µA	0.0200 µA
190.000 µA	190.000 µA	190.0009 µA	2.3E-009 A	189.9515 µA	190.0485 µA
-190.000 µA	-190.000 µA	-190.0036 µA	2.3E-009 A	-190.0485 µA	-189.9515 µA
329.000 µA	329.000 µA	328.9998 µA	7.1E-009 A	328.9307 µA	329.0693 µA
-329.000 µA	-329.000 µA	-329.0032 µA	7.1E-009 A	-329.0693 µA	-328.9307 µA

3.3 mA Range

1 Year Specification

0.00000 mA	0.00000 mA	-0.000006 mA	9.2E-010 A	-0.000050 mA	0.000050 mA
1.90000 mA	1.90000 mA	1.899999 mA	2.2E-008 A	1.899760 mA	1.900240 mA
-1.90000 mA	-1.90000 mA	-1.900012 mA	2.2E-008 A	-1.900240 mA	-1.899760 mA
3.29000 mA	3.29000 mA	3.290006 mA	8.6E-008 A	3.289621 mA	3.290379 mA
-3.29000 mA	-3.29000 mA	-3.290020 mA	8.6E-008 A	-3.290379 mA	-3.289621 mA

33 mA Range

1 Year Specification

0.0000 mA	0.0000 mA	-0.00002 mA	7.5E-009 A	-0.00025 mA	0.00025 mA
19.0000 mA	19.0000 mA	19.00002 mA	5.8E-007 A	18.99785 mA	19.00215 mA
-19.0000 mA	-19.0000 mA	-18.99998 mA	5.9E-007 A	-19.00215 mA	-18.99785 mA
32.9000 mA	32.9000 mA	32.90002 mA	1E-006 A	32.89646 mA	32.90354 mA
-32.9000 mA	-32.9000 mA	-32.90001 mA	1E-006 A	-32.90354 mA	-32.89646 mA

330 mA Range

1 Year Specification

0.000 mA	0.000 mA	-0.0002 mA	1.1E-007 A	-0.0025 mA	0.0025 mA
190.000 mA	190.000 mA	190.0006 mA	6E-006 A	189.9785 mA	190.0215 mA
-190.000 mA	-190.000 mA	-190.0006 mA	6E-006 A	-190.0215 mA	-189.9785 mA

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Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
329.000 mA	329.000 mA	329.0012 mA	9.9E-006 A	328.9646 mA	329.0354 mA	
-329.000 mA	-329.000 mA	-329.0007 mA	9.8E-006 A	-329.0354 mA	-328.9646 mA	
1.1 A Range						
1 Year Specification						
0.00000 A	0.00000 A	-0.000019 A	2.2E-006 A	-0.000040 A	0.000040 A	
1.09000 A	1.09000 A	1.089992 A	3.8E-005 A	1.089742 A	1.090258 A	
-1.09000 A	-1.09000 A	-1.090020 A	3.8E-005 A	-1.090258 A	-1.089742 A	
3 A Range						
1 Year Specification						
2.99000 A	2.99000 A	2.989966 A	0.00011 A	2.988824 A	2.991176 A	
-2.99000 A	-2.99000 A	-2.989979 A	0.00011 A	-2.991176 A	-2.988824 A	
11 A Range						
1 Year Specification						
0.0000 A	0.0000 A	-0.00009 A	1.3E-005 A	-0.00050 A	0.00050 A	
10.9000 A	10.9000 A	10.89958 A	0.00077 A	10.89405 A	10.90595 A	
-10.9000 A	-10.9000 A	-10.89957 A	0.00077 A	-10.90595 A	-10.89405 A	
20.5 A Range						
1 Year Specification						
20.0000 A	20.0000 A	19.99843 A	0.0014 A	19.97925 A	20.02075 A	
-20.0000 A	-20.0000 A	-19.99901 A	0.0014 A	-20.02075 A	-19.97925 A	

AC Current Accuracy Test

330 µA Range

1 Year Specification

33.00 uA 1 kHz	33.00 µA	33.009 µA	3.5E-008 A	32.859 µA	33.141 µA
33.00 uA 10 kHz	33.00 µA	33.008 µA	3.5E-008 A	32.536 µA	33.464 µA
33.00 uA 30 kHz	33.00 µA	33.071 µA	1.8E-007 A	32.072 µA	33.928 µA
190.00 uA 1 kHz	190.00 µA	190.053 µA	1.1E-007 A	189.662 µA	190.338 µA
190.00 uA 10 kHz	190.00 µA	189.991 µA	1.1E-007 A	188.280 µA	191.720 µA
190.00 uA 30 kHz	190.00 µA	190.280 µA	9.4E-007 A	186.560 µA	193.440 µA
329.00 uA 1 kHz	329.00 µA	329.060 µA	4E-008 A	328.489 µA	329.511 µA
329.00 uA 10 kHz	329.00 µA	329.027 µA	3.3E-007 A	326.168 µA	331.832 µA
329.00 uA 30 kHz	329.00 µA	329.043 µA	1.9E-006 A	323.336 µA	334.664 µA

3.3 mA Range

1 Year Specification

0.33000 mA 1 kHz	0.33000 mA	0.329968 mA	1E-007 A	0.329520 mA	0.330480 mA
0.33000 mA 5 kHz	0.33000 mA	0.329974 mA	2.8E-007 A	0.329140 mA	0.330860 mA
0.33000 mA 30 kHz	0.33000 mA	0.329536 mA	1E-006 A	0.326100 mA	0.333900 mA

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Certificate Number: EVL1056908

Date of Calibration: 29 May 2025

Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error Lower Limit	Upper Limit	Status
1.90000 mA 1 kHz	1.90000 mA	1.900015 mA	2.7E-007 A	1.897950 mA	1.902050 mA	
1.90000 mA 10 kHz	1.90000 mA	1.899893 mA	1E-007 A	1.890200 mA	1.909800 mA	
1.90000 mA 30 kHz	1.90000 mA	1.899742 mA	1.5E-007 A	1.880400 mA	1.919600 mA	
3.29000 mA 10 Hz	3.29000 mA	3.289133 mA	5.6E-007 A	3.283270 mA	3.296730 mA	
3.29000 mA 45 Hz	3.29000 mA	3.290022 mA	1.6E-007 A	3.286560 mA	3.293440 mA	
3.29000 mA 1 kHz	3.29000 mA	3.290045 mA	1.8E-007 A	3.286560 mA	3.293440 mA	
3.29000 mA 5 kHz	3.29000 mA	3.290100 mA	1.6E-007 A	3.283220 mA	3.296780 mA	
3.29000 mA 10 kHz	3.29000 mA	3.290075 mA	1.6E-007 A	3.273250 mA	3.306750 mA	
3.29000 mA 30 kHz	3.29000 mA	3.290246 mA	2E-007 A	3.256500 mA	3.323500 mA	

33 mA Range

1 Year Specification

3.3000 mA 1 kHz	3.3000 mA	3.29974 mA	1.4E-006 A	3.29668 mA	3.30332 mA	
3.3000 mA 5 kHz	3.3000 mA	3.30034 mA	1.4E-006 A	3.29536 mA	3.30464 mA	
3.3000 mA 30 kHz	3.3000 mA	3.30434 mA	5.9E-007 A	3.28280 mA	3.31720 mA	
19.0000 mA 1 kHz	19.0000 mA	18.99991 mA	3.2E-006 A	18.99040 mA	19.00960 mA	
19.0000 mA 10 kHz	19.0000 mA	19.00043 mA	8.4E-007 A	18.95900 mA	19.04100 mA	
19.0000 mA 30 kHz	19.0000 mA	19.00235 mA	1.3E-006 A	18.92000 mA	19.08000 mA	
32.9000 mA 10 Hz	32.9000 mA	32.89191 mA	5.6E-006 A	32.83878 mA	32.96122 mA	
32.9000 mA 45 Hz	32.9000 mA	32.89958 mA	1.4E-006 A	32.88484 mA	32.91516 mA	
32.9000 mA 1 kHz	32.9000 mA	32.89955 mA	1.4E-006 A	32.88484 mA	32.91516 mA	
32.9000 mA 5 kHz	32.9000 mA	32.89946 mA	1.4E-006 A	32.87168 mA	32.92832 mA	
32.9000 mA 10 kHz	32.9000 mA	32.89987 mA	1.6E-006 A	32.83120 mA	32.96880 mA	
32.9000 mA 30 kHz	32.9000 mA	32.90348 mA	2.7E-006 A	32.76440 mA	33.03560 mA	

330 mA Range

1 Year Specification

33.000 mA 1 kHz	33.000 mA	32.9999 mA	1.6E-005 A	32.9668 mA	33.0332 mA	
33.000 mA 5 kHz	33.000 mA	33.0045 mA	1.5E-005 A	32.9170 mA	33.0830 mA	
33.000 mA 30 kHz	33.000 mA	33.0393 mA	2.3E-006 A	32.6680 mA	33.3320 mA	
190.000 mA 1 kHz	190.000 mA	190.0030 mA	3.2E-005 A	189.9040 mA	190.0960 mA	
190.000 mA 10 kHz	190.000 mA	190.0060 mA	1E-005 A	189.5200 mA	190.4800 mA	
190.000 mA 30 kHz	190.000 mA	189.9958 mA	1.1E-005 A	189.0400 mA	190.9600 mA	
329.000 mA 10 Hz	329.000 mA	328.9183 mA	5.7E-005 A	328.3878 mA	329.6122 mA	
329.000 mA 45 Hz	329.000 mA	328.9982 mA	1.5E-005 A	328.8484 mA	329.1516 mA	
329.000 mA 1 kHz	329.000 mA	328.9986 mA	1.5E-005 A	328.8484 mA	329.1516 mA	
329.000 mA 5 kHz	329.000 mA	328.9977 mA	1.5E-005 A	328.6210 mA	329.3790 mA	
329.000 mA 10 kHz	329.000 mA	329.0004 mA	1.4E-005 A	328.2420 mA	329.7580 mA	
329.000 mA 30 kHz	329.000 mA	329.0221 mA	1.8E-005 A	327.4840 mA	330.5160 mA	

1.1 A Range

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Certificate Number: EVL1056908

Date of Calibration: 29 May 2025

Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error		Status
				Lower Limit	Upper Limit	
1 Year Specification						
0.33000 A 1 kHz	0.33000 A	0.330022 A	0.00014 A	0.329735 A	0.330265 A	
0.33000 A 5 kHz	0.33000 A	0.330607 A	0.00023 A	0.327020 A	0.332980 A	
0.33000 A 10 kHz	0.33000 A	0.331325 A	0.00055 A	0.316750 A	0.343250 A	
1.09000 A 10 Hz	1.09000 A	1.089729 A	0.00051 A	1.087938 A	1.092062 A	
1.09000 A 45 Hz	1.09000 A	1.090004 A	0.0002 A	1.089355 A	1.090645 A	
1.09000 A 1 kHz	1.09000 A	1.090053 A	0.00021 A	1.089355 A	1.090645 A	
1.09000 A 5 kHz	1.09000 A	1.090459 A	0.00027 A	1.082460 A	1.097540 A	
1.09000 A 10 kHz	1.09000 A	1.092608 A	0.00058 A	1.057750 A	1.122250 A	
3 A Range						
1 Year Specification						
2.99000 A 10 Hz	2.99000 A	2.989230 A	0.00051 A	2.984518 A	2.995482 A	
2.99000 A 45 Hz	2.99000 A	2.989982 A	0.00016 A	2.988106 A	2.991894 A	
2.99000 A 1 kHz	2.99000 A	2.989905 A	0.00016 A	2.988106 A	2.991894 A	
2.99000 A 5 kHz	2.99000 A	2.990439 A	0.0045 A	2.971060 A	3.008940 A	
2.99000 A 10 kHz	2.99000 A	2.998554 A	0.00089 A	2.910250 A	3.069750 A	
11 A Range						
1 Year Specification						
3.3000 A 500 Hz	3.3000 A	3.30001 A	0.00049 A	3.29470 A	3.30530 A	
3.3000 A 1 kHz	3.3000 A	3.29995 A	0.00057 A	3.29470 A	3.30530 A	
3.3000 A 5 kHz	3.3000 A	3.29221 A	0.00091 A	3.19900 A	3.40100 A	
10.9000 A 45 Hz	10.9000 A	10.89974 A	0.00099 A	10.89146 A	10.90854 A	
10.9000 A 65 Hz	10.9000 A	10.90001 A	0.0011 A	10.89146 A	10.90854 A	
10.9000 A 500 Hz	10.9000 A	10.90011 A	0.0013 A	10.88710 A	10.91290 A	
10.9000 A 1 kHz	10.9000 A	10.90024 A	0.0012 A	10.88710 A	10.91290 A	
10.9000 A 5 kHz	10.9000 A	10.87321 A	0.01 A	10.57100 A	11.22900 A	
20.5 A Range						
1 Year Specification						
20.0000 A 45 Hz	20.0000 A	20.00439 A	0.0094 A	19.97100 A	20.02900 A	
20.0000 A 65 Hz	20.0000 A	20.00654 A	0.0086 A	19.97100 A	20.02900 A	
20.0000 A 500 Hz	20.0000 A	20.00737 A	0.0093 A	19.96500 A	20.03500 A	
20.0000 A 1 kHz	20.0000 A	20.00797 A	0.01 A	19.96500 A	20.03500 A	
20.0000 A 5 kHz	20.0000 A	19.96769 A	0.026 A	19.39500 A	20.60500 A	

DC Power Amplitude Accuracy Test

1 Year Specification

2.19 A 45.662 mV	2.19000 A	2.189992 A	8.6E-005 A	2.189128 A	2.190872 A
45.662 mV 2.19 A	45.662 mV	45.6616 mV	6.5E-007 V	45.6601 mV	45.6639 mV

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Certificate Number: EVL1056908

Date of Calibration: 29 May 2025

Calibration Results

Nominal Value	Reference Value	Measurement Result	Expanded Uncertainty	Limits of Error Lower Limit	Upper Limit	Status
100.0 mW 45.662 mV, 2.19 A	100.00 mW	99.999 mW	4.2E-006 W	99.960 mW	100.040 mW	
100 μ A 1000 V	100.000 μ A	99.9994 μ A	1.4E-009 A	99.9650 μ A	100.0350 μ A	
1000 V 100 μ A	1000.000 V	999.9990 V	0.0051 V	999.9805 V	1000.0195 V	
100.0 mW 1000 V, 100 μ A	100.00 mW	99.999 mW	1.5E-006 W	99.965 mW	100.035 mW	
1.0 mA 1000 V	1.00000 mA	0.999997 mA	1.3E-008 A	0.999850 mA	1.000150 mA	
1000V 1 mA	1000.000 V	1000.0000 V	0.0051 V	999.9805 V	1000.0195 V	
1.0 W 1000 V, 1 mA	1.000 W	1.0000 W	1.4E-005 W	0.9998 W	1.0002 W	
2.19 A 329 V	2.19000 A	2.189994 A	8.6E-005 A	2.189128 A	2.190872 A	
329 V 2.19 A	329.0000 V	328.99967 V	0.0022 V	328.99393 V	329.00607 V	
720.5 W 329 V, 2.19 A	720.5 W	720.51 W	0.029 W	720.21 W	720.79 W	
10.9A 45.872 mV	10.9000 A	10.89934 A	0.00078 A	10.89405 A	10.90595 A	
45.872 mV 10.9 A	45.8720 mV	45.87170 mV	6.6E-007 V	45.87008 mV	45.87392 mV	
500.0 mW 45.872 mV, 10.9 A	500.0 mW	499.97 mW	3.6E-005 W	499.73 mW	500.27 mW	

AC Power Amplitude Accuracy Test

1 Year Specification

3.3 mA 65 Hz, 300 V	3.3000 mA	3.29977 mA	4E-007 A	3.29668 mA	3.30332 mA
300.0 V 65 Hz, 3.3 mA	300.000 V	300.0024 V	0.0096 V	299.9410 V	300.0590 V
990 mW 65 Hz, 300 V, 3.3 mA, 0°	990.0 mW	989.94 mW	0.00012 W	988.99 mW	991.01 mW
3.3 mA 65 Hz, 1000 V	3.3000 mA	3.30136 mA	4E-007 A	3.29668 mA	3.30332 mA
1000.0 V 65 Hz, 3.3 mA	1000.00 V	1000.021 V	0.03 V	999.690 V	1000.310 V
1.65 W 65 Hz, 1000 V, 3.3 mA, 60°	1.650 W	1.6507 W	0.00021 W	1.6447 W	1.6553 W
33 mA 500 Hz, 1000 V	33.000 mA	32.9987 mA	3.8E-006 A	32.9668 mA	33.0332 mA
1000.0 V 500 Hz, 33 mA	1000.00 V	1000.024 V	0.03 V	999.690 V	1000.310 V
33 W 500 Hz, 1000 V, 33 mA, 0°	33.00 W	33.000 W	0.0039 W	32.965 W	33.035 W
33 mA 500 Hz, 1000 V	33.000 mA	33.0084 mA	3.8E-006 A	32.9668 mA	33.0332 mA
1000.0 V 500 Hz, 33 mA	1000.00 V	1000.023 V	0.03 V	999.690 V	1000.310 V
16.5 W 500 Hz, 1000 V, 33 mA, 60°	16.50 W	16.505 W	0.002 W	16.374 W	16.626 W
33 mA 1 kHz, 1000 V	33.000 mA	32.9967 mA	3.8E-006 A	32.9668 mA	33.0332 mA

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